Pumping Lemma and Properties of Regular Languages

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What we will discuss in the next two lectures:

1. Proving that a language is not regular.
2. Closure properties of RL.
3. Finite automata equivalence checking.
4. Minimize an automaton.

1 Pumping Lemma

1.1 Not every language is regular.

- We have proved this using the countability argument.

  We will now see concrete examples.

- Example

  $L_{\text{EQ}} = \{0^n 1^n \mid n \geq 1\}$ is not regular.

  Note that it is not the concatenation of $\{0^n \mid n \geq 1\}$ and $\{1^n \mid n \geq 1\}$.

1.2 Pumping lemma

> Let L be a regular language. Then there exists a constant n (which may depend on L) such that for every w in L such that $|w| \geq n$, we can break w into three strings x, y, z as $w = xyz$, such that:
>
> 1. $y \neq \epsilon$.
> 2. $|xy| \leq n$.
> 3. For all $k \geq 0$, $xy^k z$ is also in L.

- Proof

  Proof. Use the pigeonhole principle to argue that one has to visit a state again.

  □

- It's like a game between the 'for all' player and the 'exists' player

  For all L, there is n, for all w in L, there is a partition such that for all k, ...

1.3 Examples

L_{EQ}

$L_{\text{PAL}} = \{w \mid w = w^R\}$

$L_{\text{PRIME}} = \{1^p \mid p \text{ is prime}\}$

Proof. Assume to the contrary that L_{PRIME} is a regular language.

By the pumping lemma, there is a constant n, such that for all prime $p \geq n$, we can partition 1^p into three parts x, y, z with $y \neq \epsilon$, such that $xy^k z \in L_{\text{PRIME}}$. That is, $p + (k-1)|y|$ is prime for all k. Choose $k = p+1$; we have a contradiction. So L_{PRIME} cannot be regular.

□

$L_{\text{SQUARE}} = \{1^{m^2} \mid m \in \mathbb{N}\}$

1.4 Myhill–Nerode theorem

It provides an ***if and only if*** condition for a language to be regular.

EXTRA: Check it out yourself if you are interested. Not required for this course.

2 Closure Properties

- Closure properties

  The result of certain operations on regular languages is still regular

  1. Union
  2. Closure (star)
  3. Concatenation
  4. Intersection
  5. Complement
  6. Difference
  7. Reversal
  8. Homomorphism
  9. Inverse homomorphism

2.1 Union, intersection, complement

- Union

  There are $L = L(R)$, $M = L(S)$ for RE R and S respectively. $L \cup M = L(R + S)$ by the definition of $+$.

- Complement

  Let $A = (Q, \Sigma, \delta, q_0, F)$ be the DFA for the language.

  Complement the accepting states. That is, consider $A' = (Q, \Sigma, \delta, q_0, Q \setminus F)$.

- Closure under intersection!

  Idea: Finite memory vs states

  Suppose we have DFA's A and B for languages L and M respectively.

  How to design a DFA C for $L \cap M$?

  It is known as ***the product construction***.

2.2 Reversal

- The reversal of a string $w = a_1 \cdots a_n$ is the string $a_n \cdots a_1$, denoted as w^R.

  Define $L^R = \{w^R \mid w \in L\}$.

- If L is regular, so is L^R.

  It can be proved easily via either the automata picture or the RE picture.

- See extra notes for how to ***develop a formal proof from scratch*** using the reversal as an example.